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Roll No

EC-7002-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Satellite Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Define the following parameters with reference to satellite orbits.
 - i) Apogee and Perigee
 - ii) Eccentricity
 - iii) Ascending and descending nodes.
- b) Discuss the Kepler's three laws of planetary motion.
2. a) Calculate the radius of a circular orbit for which the period is one day.
- b) Describe briefly the main effects of the earth's equatorial bulge on a satellite orbit.
3. a) Discuss about the geostationary orbit. Discuss about its limits of visibility.
- b) Discuss about the earth's launching orbits. Discuss about the sun transit outage.
4. a) Explain briefly about transponder.
- b) Explain about the receive only home TV system.

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5.
 - a) Briefly describe the equipment sections making up a transponder channel.
 - b) With the aid of block schematic, briefly describe the functioning of the indoor receiving unit of a satellite TV/FM receiving system intended for home reception.
6.
 - a) With the help of block diagram explain the operation of DBS system.
 - b) What is VSAT system? How does a VSAT work? Who uses VSAT system?
7.
 - a) Explain what is meant by EIRP. A satellite downlink at 12 GHz operates with a transmit power of 6W and an antenna gain of 48.2 dB. Calculate the EIRP in dBW.
 - b) Derive the general link equation. Find out expression for C/N and G/T ratio. Explain the importance of these ratio's in satellite link design.
8. Write down short notes on following:
 - i) Antenna look angles
 - ii) Ionospheric depolarization
